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The structure of liquid ethanol: A neutron diffraction and molecular dynamics study

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A comprehensive neutron scattering study has been performed using selective hydrogen–deuterium substitution on the hydroxyl, methyl, and methylene hydrogen atoms in pure ethanol at room temperature. In this experiment we have directly measured 10 structure factors of the possible 21 partial structure factors that exist for liquid ethanol. The data has been used to obtain details on the conformation of the liquid ethanol molecule in the liquid state. The intermolecular structure were compared to molecular dynamics simulations using the Jorgensen potential for four- and nine-site liquid ethanol molecules. All ten neutron structure factors are in good agreement with the simulation results, although there are some small differences in the structural detail. The neutron structure factors show that hydrogen bonding dominates the structure of liquid ethanol at room temperature, which is consistent with the existence of winding chains predicted by the simulation. © *2000 American Institute of Physics.* [S0021-9606(00)50813-7]

I. INTRODUCTION

The pattern of hydrogen bonding in liquid alcohols is generally very different to that found in water. For this reason simpler alcohols with only one hydroxyl group such as liquid ethanol have been the subject of many theoretical and experimental studies.[1–12] However extracting detailed structural information from experimental diffraction data on even the simplest alcohol molecules such as liquid methanol and ethanol can be quite complex. In ethanol, for example, there are six independent types of atoms yielding 21 partial structure factors. X-ray diffraction has provided useful information on the relative positions of the heavier oxygen and carbon atoms while complimentary total neutron diffraction measurements also show significant contributions from the hydrogen atoms.

Molecular dynamics and Monte Carlo techniques are well suited to the study of smaller liquid alcohols because they are economical to simulate, and much effort has been put into the development of realistic interatomic potentials.[1–3] The computer simulation results propose the existence of hydrogen bonded winding chains, a similar but much more disordered picture than that observed in the crystal structure. However, a conflicting model of distinct planar hexamer clusters have also been suggested in the liquid state at room temperature.[4] Recent developments in neutron diffraction using the hydrogen/deuterium (H/D) substitution technique[13] can now provide more detailed structural information to address these problems. For example, recent H/D substitution experiments on liquid and supercritical methanol have shown that methanol has a chain structure at room temperature but forms tetramers in the supercritical state.[14]

In this study we apply the technique of H/D substitution in neutron diffraction to liquid ethanol to obtain six intermolecular hydrogen partial structure factors corresponding to hydrogen–hydrogen interactions. In addition, we have extracted four composite structure factors. The measured neutron structure factors are compared to predictions from molecular dynamics simulation results performed using Jorgensens' optimized potential for liquid simulations[2] (OPLS) with four- and nine-site models. Particular attention is paid to the correlations between the atoms involved in the hydrogen bonding between molecules that is expected to dominate the overall intermolecular structure of the liquid.

II. PREVIOUS STUDIES

The crystal structure of ethanol[5] comprised of infinitely long zig-zag chains joined by O–H···O bonds of length 2.716 Å. The molecular conformation of ethanol has been studied in the gaseous state using microwave methods[6] and to a lesser extent in the liquid state using neutron and x-ray diffraction. Early x-ray diffraction studies on liquid ethanol were interpreted based solely on the crystal structure [7] and in a later study by Natren and Habenschuss[8] the removal of an approximate intramolecular contribution from x-ray diffraction data revealed a well-defined intermolecular O–H···O distance of 2.8 Å. However, a quantitative analysis of the liquid structure based on this data alone is not possible. Neutron diffraction data on fully deuterated liquid ethanol performed by Montague *et al.*[9] has provided information on intramolecular hydrogen distances, but the intermolecular structure was too difficult to interpret from a single diffraction pattern. Tanaka *et al.*[10] have also performed neutron diffraction measurements on fully deuterated liquid ethanol and found similar results to those of Montague *et al.*, despite

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considerable disagreement in the shape of the two total structure factors, as pointed out by Sarkar and Joarder.[4]

Jorgensen has developed optimized transferable intermolecular potentials to investigate the structure and properties of liquid alcohols using Monte Carlo simulation.[2] The results of the simulation for liquid ethanol are consistent with the presence of winding hydrogen bonded chains with about two hydrogen bonds per molecule, as found in the crystal structure. Molecular dynamics simulations of liquid ethanol over a wide temperature range by Saiz *et al.*[11] using Jorgensens' potentials also find the characteristic hydrogen bonded chain structure. A more realistic polarizable intermolecular potential functions (PIPF) for liquid alcohols have been developed by Gao *et al.*[11] through statistical mechanical Monte Carlo simulation methods. However, the results obtained using PIPF are in good agreement with the earlier calculations of Jorgensen and x-ray data of Narten (consequently, the computationally simpler Joergensen potential was used in the present study). In contrast, Sarkar and Joarder recently proposed the existence of hexamer molecular clusters in liquid ethanol at room temperature based on a comparison with Narten's x-ray data. However, the agreement of this model with the available neutron diffraction data at the total structure factor level is poor.

III. ISOTOPIC SUBSTITUTION IN NEUTRON DIFFRACTION

A. Theoretical comments

By performing neutron diffraction experiments on hydrogenous liquid samples that are identical in every respect, except that a specific hydrogen atom, it is possible to extract information on the structural environment of those substituted atom(s).[12,15] In the case of ethanol, a hydroxyl hydrogen or group of hydrogen atoms e.g., methyl or methylene hydrogen atoms within the molecule, have been isotopically substituted for by deuterium atoms(s).

Factors that govern the feasibility of measuring the difference in scattering include the scattering length contrast between the two isotopically enriched species, their atomic fraction, the instrument stability, and count rate. The large neutron scattering length difference between hydrogen and deuterium isotopes combined with the development of modern day neutron diffractometers makes H/D substitution a powerful tool in studying the structure of hydrogenous molecular liquids. There are limits to the neutron technique, however. Small quantum effects due to differences in zero-point vibrations and hydrogen bonding effects have been measured at the 1% level between the γ-ray structure factors of fully protonated and fully deuterated methanol at room temperature.[16] A series of accurate high-energy x-ray studies have recently been undertaken to investigate these effects in more detail for a range of molecular liquids.[17] For the purpose of this neutron study, the magnitude of the quantum effects are assumed to be below the limit of the experimental error of the technique for this particular case.

The measured total interference structure factor, $F(Q)$, can be expressed as the weighted sum of partial structure factors, $H_{\alpha\beta}(Q)$.[13] These, in turn, are related by Fourier transform to the site–site radial distribution functions,

$$H_{\alpha\beta}(Q)=4\pi\rho\int r^2[g_{\alpha\beta}(r)-1]\frac{\sin Qr}{Qr}dr. \tag{1}$$

The total neutron diffraction pattern of CH_3CH_2OH is comprised of 21 independent partial structure factors. In a first-order difference, the environment around the substituted hydrogen atom can be extracted, e.g., for substitution of the hydroxyl hydrogen alone in an ethanol molecule,

$$H_{XH_1}(Q)=A.[F_{CH_3CH_2OD}(Q)-F_{CH_3CH_2OH}(Q)], \tag{2}$$

the correlations from the nonsubstituted atoms cancel yielding information on the environment of the hydroxyl hydrogen alone. In a second-order difference experiment the intermolecular hydroxyl hydrogen–hydroxl hydrogen partial structure factor can be extracted explicitly through

$$H_{H_1H_1}(Q)=B.[2.F_{CH_3CH_2OM}(Q)-F_{CH_3CH_2OH}(Q) - F_{CH_3CH_2OD}(Q)], \tag{3}$$

where M is a 50:50 mixture of CH_3CH_2OH and CH_3CH_2OD to yield the maximum scattering length contrast and A and B are normalization constants. Finally, by subtracting the weighted first- and second-order differences from the total structure factor yields information on the structural environment of the nonsubstituted atoms.

B. Experimental details

Measurements were performed on 12 isotopically different samples of liquid ethanol: Fully protonated liquid ethanol CH_3CH_2OH, four specifically deuterated samples CH_3CH_2OH, CH_3CD_2OH, CH_3CH_2OD, and CD_3CD_2OD, as well as seven mixtures of these samples. In each mixture the pure samples were combined in a 50:50 ratio to maximize the scattering length contrast. The drastic change in the shape and level of scattering of the neutron differential cross section for each of the samples is shown in [Fig. 1]. From this combination of samples using the method described above, we have measured the six hydrogen–hydrogen partial structure factors present in liquid ethanol. In addition, three composite functions describing the hydrogen–carbon and hydrogen–oxygen interactions for each of the different hydrogen atom groups (e.g., hydroxyl, methylene, and methyl hydrogens) are extracted. Lastly, we have obtained a function describing the carbon–carbon, carbon–oxygen, and oxygen–oxygen interactions that is similar (but not identical) to the measured x-ray total structure factor.

The experiments were performed on the high-intensity SANDALS diffractometer at the ISIS pulsed neuton facility. High purity (99.8+%) deuterated ethanol samples of CD_3CD_2OD, CD_3CH_2OH, CH_3CD_2OH, and CH_3CH_2OD were obtained from Aldrich Chemicals Ltd. and the fully protonated ethanol sample CH_3CH_2OH (99.7+% purity) was obtained from Hayman Ltd. The experiments on the liquid samples were contained in the same 1 mm internal thickness titanium–zirconium ''null scattering'' container of wall thickness 1 mm (each). Typical sample run times were 6 h

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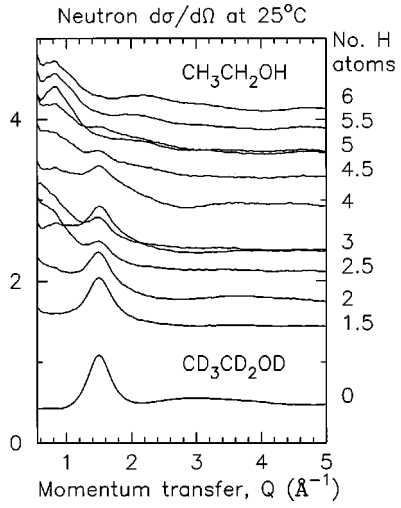


FIG. 1. The differential cross section for 12 different isotopically (hydrogen/deuterium) substituted ethanol samples, measured on the SANDALS diffractometer at an average scattering angle of $2\theta = 20.1°$. The number of hydrogen atoms per molecule is also shown to demonstrate how the differential scattering cross section changes with increasing hydrogen content.

per sample and the temperature of the sample during the experiment was 22 ± 1 °C.

C. Data analysis

The data were analyzed using the ATLAS package[18] that corrects for container scattering, attenuation, multiple scattering, nuclear recoil, and instrumental resolution. However, due to the inaccuracies of the nuclear recoil correction for hydrogen containing samples at low momentum transfer on pulsed neutron sources it was necessary to impose a constraint that the neutron structure factors, $H_{\mathrm{HH}}(Q)$, $H_{X\mathrm{H}}(Q)$, and $H_{XX}(Q)$, tended to the predicted $H(0)$ values given by the molecular dynamics simulation described in the next section. This was achieved for each dataset by Fourier transforming from $H(Q)$ to the radial distribution function $g(r)$, selecting the first physical point $g_{\mathrm{MIN}}(r)$ below which the unphysical oscillations were set to zero, and Fourier transforming back to Q space, until agreement with $H(Q \rightarrow 0)$ was reached. A minimum noise Fourier inversion technique developed by Soper[19] was used to fit the extracted neutron functions to reduce Fourier transform artifacts. It is worth noting that although the method also corrects for instrumental resolution effects, the smoothing of the neutron data to minimize Fourier transform errors appearing in real space artificially broadens out the experimental peaks in $g(r)$.

It is of importance to note that the measured $H_{XH}(Q)$ and $H_{XX}(Q)$ composite functions shown in Fig. 2 still include some intramolecular contributions from the hydrogen interactions.[13] In order to distinguish the three types of hydrogen sites in the molecule the following notation used: The hydrogen atoms on the CH_3 methyl group have been labeled H3 and those on the CH_2 methylene group H2. The hydroxl hydrogen is referred to as H1. The carbon nearest the oxygen is C1 and the carbon on the CH_3 methyl group referred to as C2. The Q-space functions in Fig. 2 contain the following contributions:

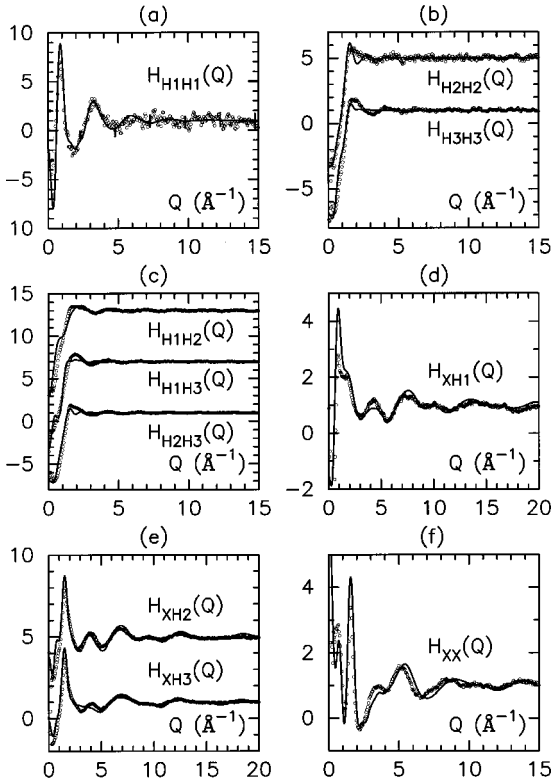


FIG. 2. The $H_{\alpha\beta}(Q)$ neutron structure factors for liquid ethanol at 22 °C. The circles represent the experimental data points and the lines are the results of the nine-site molecular dynamics simulation. The intermolecular hydrogen partial structure factors are as follows: (a) hydroxyl–hydroxyl (with some typical error bars); (b) methylene–methylene (top, shifted by +5) and methyl–methyl; (c) hydroxyl–methylene (top, shifted by +13) hydroxyl–methyl (middle, shifted by +7) and methylene–methyl. The composite structure factors including inter- and intramolecular contributions are (d) hydroxyl–nonhydrogen atoms; (e) methylene–nonhydrogen (top, shifted by +5), methyl–nonhydrogen; and (f) nonhydrogen–nonhydrogen interactions.

$$
\begin{aligned}
H_{X\mathrm{H1}}(Q) &= 0.153 H_{\mathrm{H1O}}(Q) + 0.176 H_{\mathrm{H1C1}}(Q) \\
&\quad + 0.176 H_{\mathrm{H1C2}}(Q) - 0.198 H^{\mathrm{intra}}_{\mathrm{H1H2}}(Q) \\
&\quad - 0.297 H^{\mathrm{intra}}_{\mathrm{H1H3}}(Q), \\
H_{X\mathrm{H_2}}(Q) &= 0.170 H_{\mathrm{H2O}}(Q) + 0.195 H_{\mathrm{H2C1}}(Q) \\
&\quad + 0.195 H_{\mathrm{H2C2}}(Q) - 0.110 H^{\mathrm{intra}}_{\mathrm{H2H1}}(Q) \\
&\quad - 0.330 H^{\mathrm{intra}}_{\mathrm{H2H3}}(Q), \qquad (4) \\
H_{X\mathrm{H_3}}(Q) &= 0.7369 H_{\mathrm{H3O}}(Q) + 0.8439 H_{\mathrm{H3C1}}(Q) \\
&\quad + 0.8439 H_{\mathrm{H3C2}}(Q) - 0.4749 H^{\mathrm{intra}}_{\mathrm{H3H1}}(Q) \\
&\quad - 0.9498 H^{\mathrm{intra}}_{\mathrm{H3H2}}(Q).
\end{aligned}
$$

The fourth composite neutron function $g_{XX}(r)$ is comprised of a combination of the partials involving the carbon and oxygen interactions:

$$
\begin{aligned}
H_{XX}(Q) &= 0.0924 H_{\mathrm{OO}}(Q) + 0.212 H_{\mathrm{C1O}}(Q) \\
&\quad + 0.121 H_{\mathrm{C1C1}}(Q) + 0.212 H_{\mathrm{C2O}}(Q) \\
&\quad + 0.242 H_{\mathrm{C2C1}}(Q) + 0.121 H_{\mathrm{C2C2}}(Q). \qquad (5)
\end{aligned}
$$

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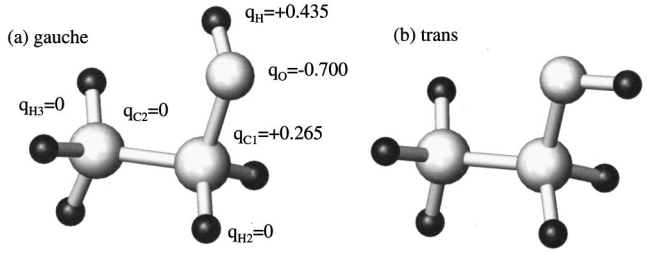


FIG. 3. Schematic diagram of an ethanol molecule in the (a) *gauche* and (b) *trans* configuration. The hydrogen substituted atoms are shown in black and the carbon and oxygen atoms in white. Also shown are the distribution of partial charges used to represent the ethanol molecule in the nine-site molecular dynamics simulation.

The total x-ray structure factor is the sum over all the scattering sites in the molecule, but for ethanol it is dominated by the C and O contributions. As such, it has many similarities with the *XX* composite neutron function,

$$H_{\mathrm{x\ ray}}(Q)=\left(\sum\ (2-\partial_{\alpha\beta})f_\alpha(Q)f_\beta(Q)H_{\alpha\beta}(Q)\right)\times\left(\sum\ f_\alpha(Q)\right)^{-2}, \tag{6}$$

where f_α is the x-ray form factor for site α.

IV. MOLECULAR DYNAMICS SIMULATIONS

Molecular dynamics simulations were carried out on 125 semiflexible molecules in cubic box of length 23 Å with periodic boundary conditions in a similar manner to Saiz *et al.*[11] The calculations were performed using the DL_POLY 2.0 molecular dynamics package[20] in the *N*, *V*, *T* ensemble. A Hoover–Nosé thermostat was used to control the temperature at $T=300$ K with a density of 0.7873 g cm^{-3}. Simulation runs of length 100 ps were carried out in steps of 0.002 ps. Two models were simulated a four-site and a nine-site model (see Fig. 3).

In the four-site model, Jorgensens' OPLS potentials were used.[2,11] Here the LJ interactions were only defined for the O–O, C–C and O–C interactions, given in Table I and truncated at 10 Å. An Ewald summation was used for Coulomb interactions. Partial charges were placed on the C1, O, and H atoms only and a dihedral potential, obtained from *ab initio* molecular orbital calculations with a 4-31 G basis set,[1] were used to describe the internal rotations about the C–O bond. The molecules were allowed to adopt either a *trans* or *gauche* configuration.

TABLE I. Lennard Jones Parameters used in molecular dynamics simulations.

Atom pair	ε (kcal/mol)	σ (Å)
O–O	0.170	3.07
C1–C1	0.110	3.98
C2–C2	0.180	3.86
O–C1	0.137	3.50
O–C2	0.175	3.44
C1–C2	0.141	3.92

TABLE II. Dihedral angle potentials used in four- and nine-site ethanol models.

Dihedral angle	V_0	V_1	V_2	V_3 (kcal/mol)
H–O–C1–C2 (four site)	0.00	0.834	−0.116	0.747
H–O–C1–C2 (nine site)	0.00	0.317	−0.176	1.378
O–C1–C2–H2 (nine site)	0.00	0.0	0.0	3.650

A nine-site model was also simulated in order to take into account the contributions from all the methyl and methylene hydrogen atoms, to which the neutron spectra are very sensitive. The simulation was essentially the same as the four-site model with the following changes: Harmonic bonds were used to mimic zero-point vibrations of the atoms in $g(r)$, all had bond strengths of 350 kcal mol^{-1} Å^{-2}. Harmonic valence angle potentials were used to define the molecular conformation of the methyl hydrogen atoms, all with a spring constant of 150 kcal mol^{-1} rad^{-2}. Noninteracting methyl and methylene hydrogens were included with zero charges (see Fig. 3) and two different dihedral potentials, derived from *ab initio* 6-31 G calculations, Gao *et al.*,[12] see Table II. These potentials were used to restrict the movement of both the hydroxyl and methyl groups about the C–O and C–C bonds (inclusion of the C–C dihedral potential was previously suggested by Saiz *et al.*,[11] who assumed only eclipsed and staggered conformations in their simulation).

V. RESULTS AND DISCUSSION

A. The molecular conformation of liquid ethanol at 22 °C

The measured Q-space partial and composite neutron structure factors are compared to the corresponding functions obtained through the nine-site molecular dynamics simulation in Fig. 2. After Fourier transformation into real space, the intramolecular contributions to the *X*H and *XX* functions were fitted to obtain the molecular conformation of the ethanol molecule in the liquid state and subtracted from the real space datasets $g_{X\mathrm{H}}(r)$ and $g_{XX}(r)$ to leave the intermolecular contributions. The intramolecular term $H^{\mathrm{intra}}_{\alpha\beta}(Q)$ was defined as

$$H^{\mathrm{int}\,ra}_{\alpha\beta}(Q)=\sum\ \frac{\sin Qr_{\alpha\beta}}{\sin Qr_{\alpha\beta}}e^{-Q^2\sigma^2/2}, \tag{7}$$

where $r_{\alpha\beta}$ is the average separation of atoms of type α and β within the molecule and σ is the standard deviation of this distance.

Overall the molecular conformation of liquid ethanol was found to be similar to the gas phase on which the simulation parameters were based (see Table III), with the intramolecular hydrogen bond lengths generally longer than those found in the crystal. As may be expected in a hydrogen bonded liquid, the intramolecular OH bond length is slightly longer than that found in the gas phase, which is consistent with limited data from previous neutron total structure factor experiments[9] on fully deuterated liquid ethanol. In addition, the current neutron data suggests that the C_1OH and C_2C_1O

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TABLE III. Ethanol molecule bond lengths.

Intramolecular bond	Crystal structure bond lengths (Å)	Jorgensen (Ref. 2) model based on gas phase data of Sasada *et al.* (Ref. 6) (Å)	Neutron liquid data of Montague *et al.* (Ref. 9) (Å)	Neutron diffraction on the liquid state (this study)	
				r (Å)	σ (Å)
H1–O	0.79(4) *trans* 0.85(3) *gauche*	0.945(2)	0.966(10)	0.97(1)	0.09
H2–C1	0.96–1.04	1.094	···	1.10(1)	0.09
H3–C2	0.93–1.05	1.094(5)	···	1.09(1)	0.09
O–C1	1.42–1.43	1.425(3)	···	1.44(1)	0.10
C1–C2	1.50–1.51	1.530(2)	1.571(16)	1.54(1)	0.10

angles are larger than in the crystal and gas phases (see Table IV), indicating that the backbone of the molecule is slightly more stretched in the liquid state.

Desipte the level of detail obtained in this experiment, some assumptions were necessary in the removal of the intramolecular scattering. Previous simulation studies[2,11] predict an equal mixture of *trans* and *gauche* configurations of the ethanol molecule in the liquid state at 25 °C. This ratio was used to separate the intra- and intermolecular contributions involving the hydroxyl hydrogen, as it was not clear from the neutron data. However, it was found that the neutron data were consistent with the methyl and methylene hydrogens orientated, on average, in the eclipsed configuration, as found in the crystal structure[5] (see Fig. 3). Consequently, this configuration was used to remove the intramolecular methyl and methylene contributions, to obtain the neutron intermolecular $g_{XH}(r)$ and $g_{XX}(r)$ functions. For the $g_{XH}(r)$ functions the intra- and intermolecular contributions were well defined with relatively small overlap, and the separation of the terms was believed to be quite accurate. However, the $g_{XX}(r)$ function in this experiment includes a contribution from all the intramolecular hydrogen–hydrogen atom interactions that is quite broad and may have lead to some systematic errors in the extracted real space function. The intermolecular Q space datasets are shown in Fig. 4. We also note that the quantum effect of zero-point vibrations of the atoms mimicked in the classical simulation by the use of a spring constant gave realistic intra- and intermolecular atomic distributions.

TABLE IV. Ethanol molecule bond angles.

Angles in degrees	Average crystal structure (Ref. 5)	Nine site simulation model (Å)	Neutron liquid data of Montague *et al.* (Ref. 9) (Å)	Neutron diffraction (this study)
∠C1OH1	104.5	108.5	106.4(1.7)	111.5(2)
∠C2C1O	108.8	108.0	111.1(1.6)	113(3)
∠H2C1H2	98.3	109.5	···	97(2)
∠H2C1C2	114.3	109.5	···	113(3)
∠H2C1O	108.8	109.5	···	111(3)
∠H3C2H3	110.5	109.5	···	110(8)
∠H3C2C1	112.2	109.5	···	109(2)

B. Comparison of neutron and MD intermolecular structure

The agreement between experiment and nine-site model simulation results in Fig. 2 is generally good, especially for the $H_{\mathrm{H1H1}}(Q)$ partial structure factor, but some small discrepancies in the other functions. These are highlighted for the $H_{XX}^{\mathrm{inter}}(Q)$ function in Fig. 5, where the resulting four-site and nine-site molecular dynamics models can be compared directly with each other and the neutron data. Whilst Jorgensens' original four-site model is in very good agreement with the *XX* neutron data, the high Q oscillations in the nine-site model are shifted to higher Q values. In real space this corresponds to a 0.08 Å shift in the first peak in $g_{\mathrm{OO}}(r)$ between the two models. These differences are a direct result of the change in dihedral potentials used to describe internal rotations of the molecule in the two models. In the four-site model of Jorgensens' a single potential to describe rotations around the C–O bond is sufficient,[1] however, in the nine-site model the two dihedral potentials of Gao *et al.*[12] are required to account for rotations around both the C–O and C–C bonds. The x-ray data of Narten is shown in comparison to the *XX* neutron function in Fig. 5, together with the molecular dynamics simulation results combined with the appropriate form factors[8] using Eq. (6). Agreement between the four-

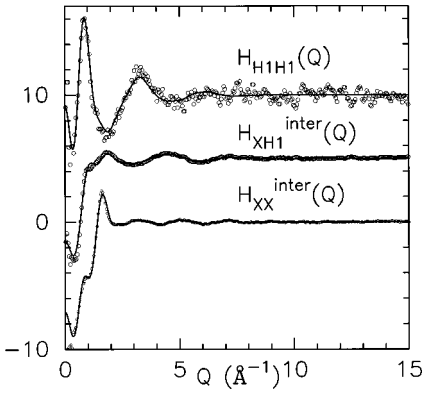


FIG. 4. The neutron intermolecular structure factors (circles) with minimum noise (Ref. 19) fits to the data (solid line): hydroxyl–hydroxyl (top, shifted by +10), hydroxyl–nonhydrogen atoms (middle, shifted by +5), and nonhydrogen–nonhydrogen interactions.

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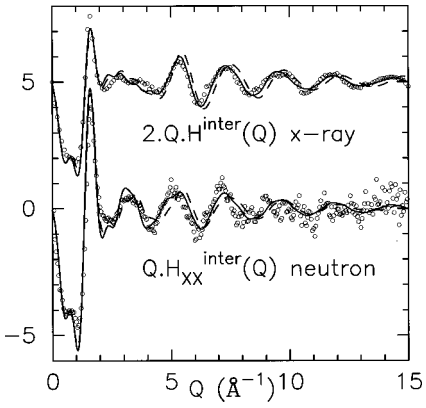


FIG. 5. A comparison of the intermolecular x-ray data of Narten and Habenschuss (Ref. 8) (circles, shifted by +5) against the similarly weighted intermolecular neutron composite structure factor (circles, bottom). Also shown are predictions of the four-site (solid lines) and nine-site (dashed lines) molecular dynamics simulations for each function.

site model and both the experimental datasets are very good. The x-ray and *XX* neutron function are both dominated by the carbon and oxygen partial structure factors. The main differences between the two functions arise from the electron form factor variation with Q. However, beyond $Q>5$ Å, where the intermolecular O–O contribution dominates, the functions have practically the same shape.

Figure 6 shows the real space maximum entropy fits of the intermolecular neutron datasets, compared to the molecular dynamics results of the four-site and nine-site models. Only the $g_{\mathrm{H1H1}}(r)$, $g_{XH1}^{\mathrm{inter}}(r)$, and $g_{XX}^{\mathrm{inter}}(r)$ functions related to the hydrogen bonding in the liquid are shown. The first peak in all three neutron distribution functions is clearly separated from the other correlations, and overall agreement between experiment and simulations is very good. However, there are some interesting differences. It has already been mentioned that there is a shift of the first peak to lower r in the $g_{\mathrm{OO}}(r)$ nine-site model compared to the four-site model. Similar shifts of 0.11 and 0.16 Å in the first peak (again to lower r in the nine-site model) are also observed in the $g_{\mathrm{H1H1}}(r)$ and $g_{\mathrm{OH}_1}(r)$ functions, respectively (see Fig. 6). While the position of the first peak in the experimental $g_{\mathrm{OO}}(r)$ function agrees best with the four-site model, the peak position in $g_{\mathrm{H1H1}}(r)$ agrees better with the nine-site model. The first peak in $g_{XH_1}(r)$ lies somewhere between the two simulation models. The corresponding running coordination numbers are shown alongside these intermolecular correlation functions, over the region of the first peak in $g(r)$. It should be noted that graph (d) was obtained assuming oxygen–hydroxyl hydrogen correlations only and graph (f), assuming just oxygen–oxygen correlations. As expected, the integrated neutron coordination numbers show that the hydroxyl hydrogen is surrounded by 2.0 ± 0.2 hydroxyl atoms by 3.0 Å, and the oxygen is surrounded by 0.95 ± 0.3 hydroxyl atoms by 2.1 Å and 2.0 ± 0.2 oxygen atoms by 3.0 Å.

In a recent model of liquid ethanol, Sarkar and Joarder[4] propose that hexamer closed chains of ethanol monomers could dominate the structure at room temperature. Other models such as a U-shaped open chain model and tetramer

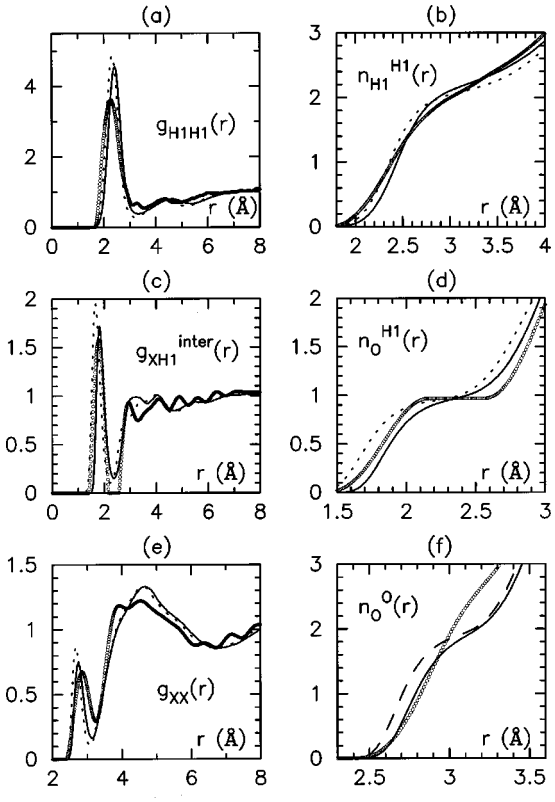


FIG. 6. The intermolecular radial distribution functions for liquid ethanol at 22 °C. In (a), (c), and (e), the circles represent the Fourier transform of the minimum noise fits to the experimental data shown in Fig. 4. Graphs (b), (d), and (f) are the corresponding running coordination numbers assuming H1–H1, O–H1, and O–O correlations only. The solid lines are the results of the four-site molecular dynamics simulation and the dashed lines the nine-site simulation model.

closed chain model are also discussed by these authors. Details from these three models (taken from Table II and III of Ref. 4) are compared to the most accurate neutron function $g_{\mathrm{H_1H_1}}(r)$ in Fig. 7. It is important to note that only the nearest neighbor [first peak in $g(r)$] interactions are considered in this comparison. It is clear from Fig. 7(a) that the closed chain hexamer and tetramer peaks do not agree well in position or area with the neutron data, while the U-shaped chain agreement is quite good. This is consistent with the results of the MD simulation for liquid ethanol and the presence of a predominantly winding hydrogen bonded chain structure, illustrated in Fig. 8. Nonetheless, a recent neutron study by Yamaguchi *et al.*[14] has shown that, on average, closed chain tetramers dominate the structure of liquid methanol in the supercritical state, and it is likely that similar structures exist in ethanol under these extreme conditions. Such experiments are in progress. Eventually, it would be intriguing to compare these neutron data against simulations performed using polarizable potentials at both ambient and supercritical conditions.

It is interesting to compare the results to those for liquid methanol.[21] Figure 7(b) compares the neutron $g_{\mathrm{H1H1}}(r)$ partial radial distribution functions for methanol and ethanol, both obtained using the same Fourier inversion procedure. As expected, the coordination number of the first peak in $g_{\mathrm{H1H1}}(r)$ is the same for methanol and ethanol within experi-

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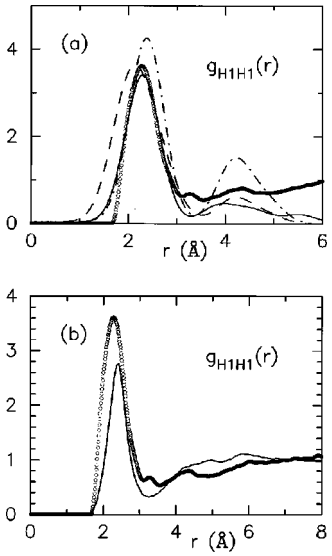


FIG. 7. (a) The measured intermolecular radial distribution function of the hydroxl hydrogen–hydroxyl hydrogen correlations, $g_{H1H1}(r)$ (circles) is shown compared to three possible models proposed by Sarker and Joarder (Ref. 4). The nearest neighbor interactions from a U-shaped chain (solid line), a tetramer closed chain (dashes), and a hexamer closed chain (dash–dot). (b) $g_{H1H1}(r)$ for ethanol (circles) compared to $g_{H1H1}(r)$ for methanol (line).

mental errors, but at a slightly shorter distance in ethanol (2.28 Å compared to 2.42 Å). In addition, there are two small peaks on the high r side of the ethanol curve, around 3.3 and 4.4 Å. The higher of these two peaks is reproduced by the ethanol molecular dynamics simulations (Fig. 5) and corresponds to the repeat H1···H1···H1 distance in the winding chain structure.

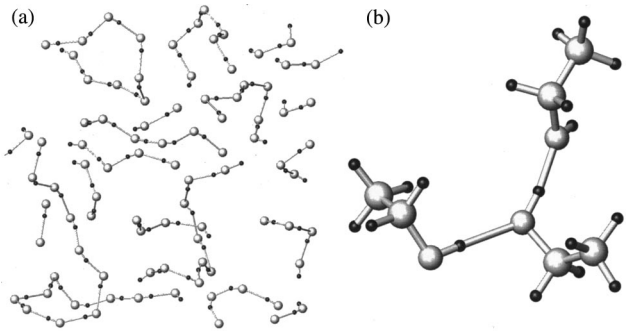


FIG. 8. Three-dimensional snapshots of the molecular dynamics simulation results for liquid ethanol at 22 °C. (a) Shows the hydrogen bonded winding chains that dominate the liquid structure. The small black circles represent the hydroxyl–hydrogen atoms and the white circles the oxygen atoms (all other atoms and some chains have been omitted for clarity). The short O–H intramolecular bond distance $r_{OH}^{intra}=0.945$ Å is shown along with longer lines $r_{OH}^{inter}<2.6$ Å, which represent the intermolecular hydrogen bonded distances. (b) In this representation three complete hydrogen bonded molecules are shown.

VI. CONCLUSIONS

A comprehensive neutron scattering study using the H/D isotopic substitution technique has been carried out on liquid ethanol at room temperature. The ability to specifically deuterate different atoms or atomic groups within the molecule without significantly altering the structure of the liquid makes the neutron technique extremely powerful in structure determination. Of the 21 partial structure factors that fully describe the structure of liquid ethanol, 6 hydrogen–hydrogen intermolecular partial structure factors have been measured as well as 4 composite neutron functions. On the simulation side, it is important to note molecular models are required that include all the hydrogen atom sites, in order to be able to compare directly with the majority of neutron scattering data. The site–site radial distribution functions describing the hydrogen bonding between molecules in the liquid compare well with molecular dynamics simulations obtained using Jorgensens intermolecular potentials.

Overall, the neutron and MD results confirm previous simulation predictions that ethanol is a strongly hydrogen bonded liquid consisting predominantly of winding chains, similar to methanol. While closed chain hexamer clusters may survive in small quantities within the liquid at room temperature, it would be interesting to see if there is a transition from a winding chain structure to clusters (at higher pressures and temperatures) as observed in the case of supercritical methanol.[14]

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